

JUNIPERO VERBEKE

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Electrical Engineering M.S. candidate with a B.S. in Applied Physics, focused on **embedded systems, control theory, and dynamic modeling**, with hands-on experience in **MATLAB/Simulink, Python, and AI/ML** for anomaly detection and yield optimization. My passion is hard tech: building and testing real hardware—from flight controller tuning to sensor integration—and **turning models into robust control software**.

American, French and Belgian citizen based in San Jose, CA, interested in internships from May 11th to August 17th, 2026, with mobility to US and European regions.

EDUCATION

San Jose State University — <i>M.S. Electrical Engineering</i>	Expected June 2027
University of California, Santa Cruz — <i>B.S. Applied Physics</i>	June 2025
Etoile du Matin - Eguelshardt, France — <i>French Baccalaureate - Mention très bien</i>	June 2020

EXPERIENCE

Custom Mechatronic Systems Design & Fabrication <i>Personal Business; CAD, Controls</i>	June 2020 – Present
- Designed and fabricated custom mechanical and electronic components for personal and client projects using 3D printing, CNC, and water jet cutting. - Built, programmed, and tuned 4 FPV drones, integrating flight controllers, sensors, and power distribution systems for personal videography business.	
Veritas Managed Solutions, Inc. <i>Applied Engineer</i>	June 2021 – Present
- Engineered and estimated over 200 security projects, including large-scale deployments at Tesla Gigafactories in Sparks and Austin. - Programmed and configured wired security cameras, card access systems, and electronic door locks. - Spearheaded the formation of the engineering team, enabling new lines of business and contributing to revenue growth in the security sector from \$10M to \$28M. - Authored 5 universal guides for technician system implementation, reducing deployment time by 5%. - Automated proposal generation using Word, Visio, and Excel, decreasing creation time by 80%.	
St. Thomas More School <i>Technology Support</i>	Sept. 2022 – Present
Maintained the school's computer network and media equipment.	

PROJECTS

Real-Time LED Backlighting with Arduino <i>Arduino, C++, Embedded Systems</i>	Dec 2023
- Developed an ambient lighting system using an Arduino microcontroller to control digital LEDs for real-time screen backlighting. - Used Arduino libraries to process screen color data and drive the LED strips, enhancing user experience.	
Physics-Based Numerical Modeling of Dynamic Systems <i>Python, NumPy, Matplotlib, MATLAB</i>	Nov. 2023
- Developed physics-based numerical models for complex dynamic systems from first principles using Lagrangian mechanics. - Analyzed and compared the computational efficiency of RK4 and Euler numerical methods in Python and MATLAB.	
Aerodynamic Analysis and Fluid System Validation <i>XFLR5, Data Analysis</i>	March 2025
Collected and analyzed experimental fluid system data, validating results against theoretical principles and XFLR5 simulations.	

SKILLS

Software & Languages: Python (NumPy, Matplotlib, SciPy), MATLAB/Simulink, Git/GitHub, VS Code, Unix Shell
Hardware & Fabrication: FPV Drone Design and Assembly, Arduino, Flight Controller Configuration, Sensor Integration, 3D Printing, CNC, Water Jet Cutting
EE Concepts & Coursework: Dynamic Modeling, Control Systems (Design, Analysis, PID Tuning), Linear Control Analysis, Applied Mathematics (Linear Algebra, Vector Calculus, Differential Equations), Circuit Analysis